RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 22/05/2026	**CIE – 2**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management& Economics		**Course Code : HS261TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	In the BCG matrix, what does a "Star" represent? Why is it important for a company?	02	2	CO2
2.	Differentiate Strategic and Operational plans on the basis of time horizons.	02	2	CO2
3.	Give two examples of corporate level strategy	02	2	CO2
4.	When price of a product falls from ₹20 to ₹16, quantity demanded rises from 100 units to 140 units. Find Price Elasticity of Demand.	02	3	CO3
5.	If the total cost of the basket in the base period (say January 2008) is $600, and in the current period (say February 2008) it is $616, calculate the CPI.	02	3	CO3
	Part – B			
1.	The rapid growth of electric vehicles (EVs) in India has increased the demand for skilled EV technicians and battery maintenance experts. However, the limited availability of trained workers resulted in higher wages, labor shortages, and delays in service operations. To overcome this challenge, companies and the government introduced training and skill development programs to increase the supply of skilled labor. Analyze the impact of demand and supply forces on: a) Wages of EV technicians b) Employment opportunities c) Labor shortages in the EV sector d) Skill development initiatives	10	4	CO4
2.	The café industry in Bengaluru has become highly competitive due to the increasing number of cafés offering similar products such as coffee and snacks. To attract customers, cafés differentiate themselves through ambience, branding, pricing, customer service, and promotional activities. "Urban Brew Café," which initially enjoyed high profits, has started losing customers after the entry of new competitors offering attractive pricing and unique customer experiences. As a result, the café faces challenges in maintaining profitability and customer loyalty in a monopolistically competitive market. Based on the above case, discuss the role of monopolistic competition in business performance with reference to: a) Product differentiation and its impact on profit b) Advertising strategies used to attract customers c) Pricing strategies adopted by cafés d) Techniques for business survival in a competitive market	10	4	CO4

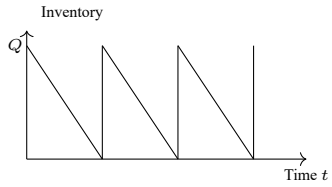
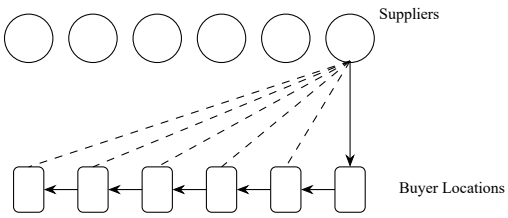
3.	a Illustrate the Oligopoly's version of the prisoner's dilemma with an example.	05	2	CO2
	b What is the expected economic impact of Budget 2026?	05	2	CO2
4.	India's electric vehicle (EV) industry is experiencing rapid growth due to government initiatives like FAME-II, rising fuel prices, and increasing environmental awareness. However, companies operating in this sector face challenges such as inadequate charging infrastructure, dependency on imported lithium-ion batteries, intense competition from domestic and global players, and rapidly changing technology. In this context, analyze how an Indian EV manufacturing company can use strategic planning to achieve sustainable growth, improve market share, and ensure long-term competitiveness in the evolving Indian automotive industry.	10	4	CO4
5.	Zeta Pharma, a Hyderabad-based pharmaceutical company supplying essential medicines across India, faced a sudden crisis in 2026 when geopolitical trade restrictions disrupted the import of a key raw material used in a life-saving drug. With only limited stock available, the company risked stopping production within weeks, affecting hospitals and patients nationwide. Since no standard procedure existed for such a disruption, top management had to take non-programmed managerial decisions under uncertainty. They evaluated options such as expensive emergency imports, prioritizing critical drug production, and collaborating with competitors or government agencies. The situation required quick judgment, risk analysis, and strategic thinking to ensure uninterrupted medicine supply. Give the analysis of the Risk analysis and strategic thinking to be adopted in the case.	10	4	CO4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 26.05.2026	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.
2. Use of Normal Distribution Tables permitted.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the inference drawn from this figure? 	2	2	CO3
2.	The primary role of cycle inventory is to allow different stages in the supply chain to purchase product in lot sizes that minimize the sum of the _____, _____, and _____ cost.	2	1	CO3
3.	A key to reducing lot size without _____ costs is to _____ the fixed cost associated with each lot.	2	1	CO2
4.	What is intermodal transportation?	2	1	CO2
5.	Identify the transportation design option that is illustrated in the sketch. List one application of this design. 	2	2	CO2
	Part - B			
1	With a help of suitable case example elucidate tailored transportation decision made based on size of customer.	10	2	CO2
2	Discuss trade-offs in transportation design with examples.	10	2	CO2
3	Demand for the Deskpro computer at Best Buy is 1,000 units per month. Best Buy incurs a fixed order placement, transportation, and receiving cost of $4,000 each time an order is placed. Each computer costs Best Buy $500 and the retailer has a holding cost of 20 percent. Evaluate the number of computers that the store manager should order in each replenishment lot, number of orders per year, annual inventory cost, and average flow time.	10	3	CO3

4	Weekly demand for HP printers at a Sam's Club store is normally distributed, with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders 1,000 printers each time the inventory drops to 600 printers. HP currently takes two weeks to fill an order. How much safety inventory does the store carry as a result of this policy? What fill rate does the store achieve? Assume that the supply lead time from HP is normally distributed, with a mean of 2 weeks and a standard deviation of 1.5 weeks. How much safety inventory should Sam's Club carry if it wants to provide a CSL of 95 percent?	10	3	CO3
5	Assume that weekly demand for phones at B&M Office Supplies is normally distributed, with a mean of 2,500 and a standard deviation of 500. The manufacturer takes two weeks to fill an order placed by the B&M manager. The store manager currently orders 10,000 phones when the inventory on hand drops to 6,000. Evaluate the safety inventory and the average inventory carried by B&M. Also evaluate the average time a phone spends at B&M. Weekly demand for Legos at a Wal-Mart store is normally distributed, with a mean of 2,500 boxes and a standard deviation of 500. The replenishment lead time is two weeks. Assuming a continuous review replenishment policy, evaluate the safety inventory that the store should carry to achieve a CSL of 90 percent.	05 05	4	CO3

Academic year 2025-2026(Even Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**25th MAY 2026**	**Maximum Marks**	**50**
Course Code	**IM363IA**	**Duration**	**90 Min**
Sem	**VI Semester**	**Test-2**	
ERGONOMICS			

PART A

S No	Questions	M	BT	CO
1	How does repetitive motion contribute to musculoskeletal disorders?	02	L2	2
2	Define moment arm in the context of biomechanics.	02	L1	3
3	List any two types of controls used in human-machine systems.	02	L2	4
4	Differentiate between discrete controls and continuous controls.	02	L2	2
5	What are labels and icons in display design?	02	L2	2

PART B

S No	Questions	M	BT	CO
1	Design an ergonomic human-machine interface for a workplace of your choice by integrating suitable displays, controls, icons, labels, and navigation aids. Justify your design decisions based on ergonomic principles.	10	L4	1
2	Analyse the ergonomic risks associated with poorly designed displays and controls in workplaces. Discuss how improper interface design can contribute to work-related musculoskeletal disorders (WMSDs).	10	L3	3
3	Explain the different types of displays used in ergonomics and discuss the major tasks performed through displays in human-machine systems. Illustrate the role of monitoring displays and integrative displays with suitable examples.	10	L3	1
4	Describe the major causes of low back problems in workplaces. Explain the ergonomic risk factors that contribute to lower back injuries among workers.	10	L2	3
5	A worker in a warehouse repeatedly lifts heavy boxes from the floor to shoulder height. Using ergonomic principles and the NIOSH lifting guide, analyse the risks involved and suggest suitable ergonomic improvements.	10	L4	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max	Quiz		06	02	02		02	08				
	Marks	Test	20	10	20				10	30	10		

Academic year 2025 - 26 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**26th May 2026**	**Maximum Marks**	**10 + 50**
Course Code	**IM364TA**	**Duration**	**120 Min**
Sem	**VI Semester**	**CIE – II**	
Human Resource Management & Analytics - IM364TA			

Part – A

S No	Questions	M	BT	CO
1.	Distinguish between *predictive validity* and *concurrent validity*.	01	L1	CO2
2.	Define *negligent hiring* in the context of HR selection.	01	L1	CO2
3.	What is meant by *validity* in employee testing?	01	L1	CO2
4.	What is an *in-basket exercise*?	01	L2	CO3
5.	Differentiate between orientation and onboarding.	01	L1	CO3
6.	What is on-the-job training?	01	L1	CO3
7.	Define appraisal interview.	01	L1	CO3
8.	Define graphic rating scale in performance appraisal	01	L2	CO3
9.	What is succession planning in talent management	01	L2	CO3
10.	A company identifies a gap between employee skills and job requirements. What is the first step in the training process?	01	L2	CO3

Part – B

S No	Questions	M	BT	CO
1.	List and describe eight types of employment tests used in employee selection. For each type, state: (i) what it measures, and (ii) the category of jobs or employees for which it is most suitable.	10	L2	CO2
2.	A multinational company is selecting candidates for a middle management role. Design a Management Assessment Centre for this role. Specify at least three simulations, the managerial competency each exercise measures, and how assessors should evaluate candidate performance in each exercise.	10	L3	CO2
3.	A bank introduced digital banking services, but employees lacked the technical skills to operate the new system effectively. Develop a suitable training program to help employees adapt to the technological changes.	10	L3	CO3
4.	A service organization plans to evaluate employee performance using feedback from supervisors, peers, subordinates, and customers. Recommend a suitable appraisal technique and explain how it can be implemented effectively.	10	L3	CO3
5.	Employees in a company often become defensive during appraisal interviews and fail to accept feedback positively. Develop guidelines for managers to conduct effective appraisal interviews that encourage employee improvement.	10	L3	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Academic year 2025 - 26 (EVEN Sem)

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	–	04	06	–	06	04	–	–	–	–
	Test	Max Marks	–	20	30	–	–	10	40	–	–	–

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DEPARTMENT OF
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Date : 15th Dec 2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS251TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Define organization structure?	1	L1	CO2
2	Define the term span of control.	1	L2	CO2
3	What does “threat of substitutes” refer to in Porters Five force Model?	1	L2	CO2
4	Who developed the concept of strategy in 1960s and is also called the father of strategic management?	1	L1	CO2
5	What are standing plans?	1	L1	CO2
6	If product passes through different stages of production, the suitable form of departmentation is _____.	1	L2	CO2
7	What is elasticity of demand?	1	L1	CO4
8	If the percentage increase in price is less than the percentage decrease in quantity demanded, the good is said to be highly _____.	1	L2	CO4
9	What do you understand by the term monopolistic competition?	1	L2	CO4
10	What does the term Equilibrium in Markets for Goods and Services mean?	1	L2	CO4
	Part – B			
1	Discuss the key differences between strategic and tactical planning.	10	L2	CO2
2	Explain the term corporate strategy. Discuss the importance of corporate strategy for a large scale business enterprise.	10	L2	CO2
3	Discuss the effectiveness of mechanistic and organic structures in different environmental conditions.	10	L3	CO2
4	Illustrate with examples the various forms of departmentalization.	10	L2	CO2
5	What are the characteristic features of an oligopolistic industry? How does it differ from monopolistic competition?	10	L2	CO4

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DEPARTMENT OF
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Date : 4th June 2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Principles of Management and Economics - HS261TA		

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Differentiate between strategic goals and operational goals.	02	2	2
2.	In the BCG matrix, what does a "Cash Cow" represent?	02	2	2
3.	List any two types of corporate strategies and provide examples.	02	1	2
4.	How does an organic structure differ from a mechanistic structure?	02	2	2
5.	If the price elasticity of demand for a product is -0.5 and the price elasticity of supply is +2.5, what do these values imply about consumer and producer responses to price changes?	02	2	4
	Part – B			
1.	Explain the strategic management process. Discuss each stage briefly and explain how this process helps organizations achieve their long-term objectives.	10	3	2
2.	A new entrant in the online grocery market wants to survive among large players like Amazon Fresh and BigBasket. a) Use Porter's Five Forces to analyze the challenges. b) Suggest a competitive strategy the company should adopt and why.	10	4	2
3.	A growing bakery business is expanding from one shop to five locations. The owner notices confusion among staff regarding their tasks, delays in decision-making, and poor coordination between branches. a) Explain the key elements of organizational structure. b) For each element, suggest practical changes the company can implement to resolve the issues the bakery. Explain how these changes will improve organizational effectiveness.	10	3	2
4.	Explain the concepts of demand and supply using appropriate diagrams. How is equilibrium price and quantity determined in a competitive market? What happens when there is a shift in either demand or supply?	10	2	4

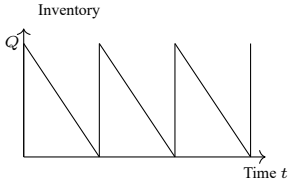
5.	Two segments of the food delivery market are being studied: • Segment A includes dozens of local restaurants offering similar menus but trying to stand out through branding, discounts, and customer experience. • Segment B includes just three major food delivery platforms that dominate logistics, pricing control, and customer data. **Questions:** a) Identify which segment represents **monopolistic competition** and which represents **oligopoly**, justifying your answer with characteristics of each market structure. b) Compare and contrast the pricing strategies and customer acquisition methods used in both segments. c) As a new entrepreneur entering the market, which segment would you prefer to compete in and why? Base your answer on market behavior, barriers to entry, and strategic flexibility.	10	4	4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 06.05.2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.
2. Use of Normal Distribution Tables permitted.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	 What is the inference drawn from the figure?	2	1	CO3
2.	The primary role of cycle inventory is to allow different stages in the supply chain to purchase product in lot sizes that minimize the sum of the _____, _____, and _____ cost.	2	1	CO3
3.	A key to reducing lot size without _____ costs is to _____ the fixed cost associated with each lot.	2	1	CO3
4.	List the Macro-economic factors influencing network design decisions.	2	1	CO2
5.	Illustrate different flexibility configurations in network.	2	1	CO2
	Part –B			
1	Develop the framework for network design decisions. Analyze the various phases.	10	3	CO2
2	Explain the methodology of evaluating network design decisions using decision trees.	10	3	CO2
3	Demand for the Deskpro computer at Best Buy is 1,000 units per month. Best Buy incurs a fixed order placement, transportation, and receiving cost of $4,000 each time an order is placed. Each computer costs Best Buy $500 and the retailer has a holding cost of 20 percent. Evaluate the number of computers that the store manager should order in each replenishment lot, number of orders per year, annual inventory cost, and average flow time.	10	3	CO3
4	Write a detailed note on multi-echelon cycle inventory.	10	2	CO3
5 a.	Assume that weekly demand for phones at B&M Office Supplies is normally distributed, with a mean of 2,500 and a standard deviation of 500. The manufacturer takes two weeks to fill an order placed by the B&M manager. The store manager currently orders 10,000 phones when the inventory on hand drops to 6,000. Evaluate the safety inventory and the average inventory carried by B&M. Also evaluate the average time a phone spends at B&M.	05	4	CO3

5 b.	Weekly demand for Legos at a Wal-Mart store is normally distributed, with a mean of 2,500 boxes and a standard deviation of 500. The replenishment lead time is two weeks. Assuming a continuous review replenishment policy, evaluate the safety inventory that the store should carry to achieve a CSL of 90 percent.	05		

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 9th June 2025	**CIE – 2**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Which type of control is most suited for precise positioning tasks?	01	L_2	CO1
2.	What is the principle of "legibility" in display design?	01	L_1	CO2
3.	What ergonomic principle recommends grouping related controls together?	01	L_2	CO2
4.	Give one example, on how you utilize the concept of minimizing information access cost to ATM screen design.	01	L_3	CO1
5.	State where a desktop monitor should be positioned for ergonomic efficiency based on anthropometric data.	01	L_2	CO2
6.	State how the redundancy gain principle is used in designing emergency alerts.	01	L.2	CO2
7.	What is the primary ergonomic concern in designing for children?	01	L_2	CO2
8.	Give one example of assistive technology.	01	L_1	CO1
9.	What is one reason to avoid "one-size-fits-all" designs?	01	L_2	CO2
10.	Name one cognitive difference relevant in the design for children.	01	L_1	CO1
	Part – B			
1.	Explain how you would apply the Principles of Display Design when designing a monitoring panel in an industrial plant.	10	L_3	CO2
2.	Compare and contrast discrete and continuous controls in terms of their ergonomic applicability with suitable examples.	10	L_3	CO2
3 a.	Briefly outline how the principles of Fitts's Law is applied to justify the optimal placement of frequently used touch controls on a tablet interface.	05	L_3	CO3
b.	A user navigates to a target 160 mm away and 40 mm wide. With a = 80 and b = 90, compute movement time.	05	L_3	CO2
4.	How can anthropometric data be used to improve the user interface of a digital product? Provide examples involving touchscreens, workstations, or ATMs.	10	L_3	CO2
5.	Discuss the special considerations to be taken into account when designing for children. Highlight how anthropometric data of children differs from that of adults.	10	L_4	CO4

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DEPARTMENT OF
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Date : 5th June 2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Human Resource Management & Analytics - IM364TA		

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Differentiate between content validity and criterion validity in test.	02	L1	CO2
2.	Define unstructured interview and structured interview.	02	L2	CO2
3.	Mention the ways to evaluate training effectiveness.	02	L2	CO3
4.	Define behaviorally anchored rating scale.	02	L1	CO3
5.	Explain reactive decision-making and predictive decision-making in HR.	02	L2	CO3
	Part – B			
1.	Elaborate on the major types of employment selection tests that organizations use during recruitment. For each category, describe its purpose, what it measures, and provide relevant job-related examples to illustrate its application in real-world hiring.	10	L2	CO2
2.	Describe the different types of interviews used in the selection process with examples and how does theses interview types help assess candidate suitability.	10	L3	CO2
3.	Imagine you are responsible for implementing a training program. Outline the systematic steps involved in designing and executing an effective training process. For each step, explain its role and importance in ensuring the training meets both organizational and employee development goals.	10	L3	CO3
4.	Identify frequent challenges or errors that reduce the effectiveness of performance appraisals. Suggest strategies or best practices that can be adopted to minimize these issues and improve the overall fairness, accuracy, and impact of the appraisal system.	10	L2	CO3
5.	Describe the framework of HR Analytics. Discuss the key components and explain how they contribute to strategic HR decision-making.	10	L2	CO3

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**07 January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**HS251TA**	**Duration**	**120 Min**
Sem	**V**	**CIE – II**	
PRINCIPLES OF MANAGEMENT AND ECONOMICS			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	_____ plans are specifically designed to meet the needs of a unique situation.	01	1	1
2	Creating or changing the organization structure is called _____.	01	1	1
3	Identify the Quadrant of the BCG Matrix which represents high market share and high market growth.	01	1	1
4	_____ planning bridges the gap between strategic goals and operational tasks.	01	1	2
5	_____ refers to how standardized an organization's jobs are and the extent to which employee behaviour is guided by rules and procedures.	01	1	1
6	A rigid, tightly controlled structure, characterized by high specialization is a characteristic of _____ organization.	01	1	2
7	Draw the circular flow diagram with reference to microeconomics.	02	1	4
8	The rate of increase in prices over a given period of time is defined as _____.	01	1	4
9	The _____ measures the change in prices paid by consumers for goods and services.	01	1	5
	Part – B			
1	Many companies have a goal of becoming more environmentally sustainable. One of the most important steps they can take is controlling E-waste. You've been put in charge of creating a program to do this for your company to develop plans. Describe your plans in terms of being: (a) Strategic and operational. (b) Short term and long term.	10	2	2
2	Bring out the ways in which a firm could be organized for smooth functioning. How the Chain of command and Span of control does affect the efficient functioning of the firm.	10	2	2
3	An organization should be viewed as a unified, purposeful system composed of interrelated parts. Illustrate using food industry as an example.	10	4	1
4	Will the firms in an oligopoly act more like a monopoly or more like competitors? Briefly explain with suitable example.	10	3	4

5	A rapidly developing country has hired you to present a specific strategy for keeping their inflation and unemployment rates low while maintaining a healthy rate of GDP growth. Explain in detail the components of GDP that can be used to accomplish this.	10	3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max	Quiz	04	02	–	03	01	10	–	–	–	–	–
	Marks	Test	10	20	–	20	–	–	20	20	10	–	–